Data Cleaning Report: Café Sales

1. Project Overview

In this project, we're gonna go through the entire process of cleaning data using Google Sheets, because the dataset isn't big enough to use another tool.
We're going to follow five steps to make sure that the dataset is clean for further analysis.

1. Initial Inspection and Data Types
2. Nulls and Whitespaces
3. Duplicates
4. Text Standardization
5. Outlinders

2. Dataset Overview & Data Quality Assessment

Primary Data Source

- Source: Kaggle (dirty_cafe_sales.csv)
- Initial Dimensions: 10,000 rows × 8 columns
- Key Columns: Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, Transaction Date

Data location

The dataset was downloaded from Kaggle and stored locally. Copies of the original files were created to ensure that the raw data remains unchanged during the cleaning process

Data Credibility (ROCCC)

- The dataset comes with a license CC BY-SA 4.0 (Creative Commons Attribution-ShareAlike 4.0 International); it guarantees free use, modificatión y públic distributión for educational and portfolio purposes
- Being a simulation dataset (synthetic), it has an explicit design for the practice of *Data Cleaning* and contains deliberate errors (nulls, syntax errors)

Limitations

As a synthetic dataset geared towards training, it lacks context such as extensive dates or metadata from real customers, so operational conclusions are limited to validating the information.

3. Step-by-Step Data Cleaning Process

Initial Inspection and Data Types:

- **Data Types Identified:** The dataset contains a mix of text, numerical values, and dates across different columns.
- **Transaction ID Analysis:** The `Transaction ID` column includes alphanumeric values (a combination of letters and numbers).
- **Next Steps:** In the upcoming data-cleaning phases, this column will be audited for duplicates to verify individual sales and ensure that each transaction corresponds correctly to its ID.

Handling Null Values, Errors, and Numeric Imputation:

1. Text & Date Fields Standardization

- **Findings:** A high volume of missing cells (blanks), #ERROR! values, and "Unknown" entries were identified across categorical and date columns (Item, Location, Transaction Date, and Payment Method).
- **Action:** To maintain consistency and avoid losing transactional records, all missing or erroneous non-numeric entries were standardized under a unified "Unknown" label.

2. Numeric Fields Imputation & Calculation

Missing values across financial and numerical columns were resolved using logical and mathematical rules:

- **Price per Unit:** Missing values were recalculated using $\text{Total Spent} / \text{Quantity}$. Where applicable, standard prices were derived directly from known product names using the product pricing catalog.
- **Quantity:** Missing values were calculated using $\text{Total Spent} / \text{Price per Unit}$.
- **Total Spent:** Calculated in a newly dedicated column using the formula $\text{Quantity} \times \text{Price per Unit}$.
- **Unresolvable Numerics:** Rows containing simultaneous missing entries across all three numeric variables (where mathematical derivation was impossible) were standardized to "Unknown".

3. Data Quality & Audit Check

- A final column-by-column audit was executed to ensure no unhandled #ERROR! values, blank cells, or unstandardized entries remained in the dataset.

Deduplication & Format Standardization

- **Deduplication Check:** Ran Google Sheets' native deduplication tool across all 10,000 records. No duplicate rows were identified, confirming that each transaction record is unique.
- **Currency Standardization:** Standardized the `Price per Unit` and `Total Spent` columns to US Currency format ($ USD) to ensure proper visual readability and accurate numeric processing.
- **Data Type & Field Formatting:** Performed a column-by-column audit to verify that all fields maintain their appropriate data types (text, numerical, and currency formats) for downstream analysis and visualization.

4. Key Takeaways & Recommendations for Data Collection

- **Observation:** Multiple records contained only a `Transaction ID` alongside partial data, leaving several adjacent fields incomplete or labeled as `"Unknown"`.
- **Decision Against Omission:** These sparse records were intentionally retained rather than deleted. Since they still represent confirmed sales activity—often retaining critical data points like transaction dates or revenue values—removing them would result in lost financial volume and biased reporting.
- **Analytical Impact & Future Risk:** While preserving these rows prevents the artificial reduction of total revenue, missing categorical data (e.g., unspecified

items or locations) may limit granular segmentations during downstream analysis. This trade-off ensures overall financial accuracy while acknowledging data completeness boundaries.